

Probability & Random Processes (EE2011)

Assignment # 1 (Total =100 points)

Section: BEE-4B

Due Date: 19th, Feb, 2025

Question No. 1 (CLO No. 1)

An engineering firm is hired to determine if certain waterways in Virginia are safe for fishing. Samples are taken from three rivers.

- (a) List the elements of a sample space S , using the letters F for safe to fish and N for not safe to fish.
- (b) List the elements of S corresponding to event E that at least two of the rivers are safe for fishing.
- (c) Define an event that has as its elements the points {FFF, NFF, FFN, NFN}

Question No. 2 (CLO No. 1)

A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is

- (a) an industrial engineering major
- (b) a civil engineering or an electrical engineering major.

Question No. 3 (CLO No. 1)

In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that

- (a) the student took mathematics or history;
- (b) the student did not take either of these subjects;
- (c) the student took history but not mathematics.

Question No. 4 (CLO No. 1)

Suppose the manufacturer's specifications for the length of a certain type of computer cable are 2000 ± 10 millimeters. In this industry, it is known that small cable is just as likely to be defective (not meeting specifications) as large cable. That is, the probability of randomly producing a cable with length exceeding 2010 millimeters is equal to the probability of producing a cable with length smaller than 1990 millimeters. The probability that the production procedure meets specifications is known to be 0.99.

- (a) What is the probability that a cable selected randomly is too large?
- (b) What is the probability that a randomly selected cable is larger than 1990 millimeters?

Question No. 5 (CLO No. 1)

Factory workers are constantly encouraged to practice zero tolerance when it comes to accidents in factories. Accidents can occur because the working environment or conditions themselves are unsafe. On the other hand, accidents can occur due to carelessness or so-called human error. In addition, the worker's shift, 7:00 A.M.–3:00 P.M. (day shift), 3:00 P.M.–11:00 P.M. (evening shift), or 11:00 P.M.–7:00 A.M. (graveyard shift), may be a factor. During the last year, 300 accidents have occurred. The percentages of the accidents for the condition combinations are as follows:

Shift	Unsafe Conditions	Human Error
Day	5%	32%
Evening	6%	25%
Graveyard	2%	30%

If an accident report is selected randomly from the 300 reports,

- (a) what is the probability that the accident occurred on the graveyard shift?
- (b) what is the probability that the accident occurred due to human error?
- (c) what is the probability that the accident occurred due to unsafe conditions?
- (d) what is the probability that the accident occurred on either the evening or the graveyard shift?

Question No. 6 (CLO No. 1)

From past experience, a stockbroker believes that under present economic conditions a customer will invest in tax-free bonds with a probability of 0.6, will invest in mutual funds with a probability of 0.3, and will invest in both tax-free bonds and mutual funds with a probability of 0.15. At this time, find the probability that a customer will invest

- (a) in either tax-free bonds or mutual funds
- (b) in neither tax-free bonds nor mutual funds.

Question No. 7 (CLO No. 1)

The probability that an American industry will locate in Shanghai, China, is 0.7, the probability that it will locate in Beijing, China, is 0.4, and the probability that it will locate in either Shanghai or Beijing or both is 0.8. What is the probability that the industry will locate

- (a) in both cities?
- (b) in neither city?

Question No. 8 (CLO No. 1)

Which of the following events are equal?

- (a) $A = \{1, 3\}$;
- (b) $B = \{x \mid x \text{ is a number on a die}\}$;
- (c) $C = \{x \mid x^2 - 4x + 3 = 0\}$;
- (d) $D = \{x \mid x \text{ is the number of heads when six coins are tossed}\}$.

Question No. 9 (CLO No. 1)

A drug for the relief of asthma can be purchased from 5 different manufacturers in liquid, tablet, or capsule form, all of which come in regular and extra strength. How many different ways can a doctor prescribe the drug for a patient suffering from asthma?

National University of Computer & Emerging Sciences, Lahore
Department of Electrical Engineering (Spring 2025)

Question No. 10 (CLO No. 1)

In how many ways can 5 starting positions on a basketball team be filled with 8 men who can play any of the positions?